EMBEDDED HACKING

Reverse Engineering the Raspberry Pi Pico 2

Operation Zero Hour: Classified Intelligence Briefing

ARM Cortex-M33 • GDB • Ghidra

October 17, 2026

For educational and defensive security research only

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+------------------------------------------------------------------+
|                       TOP SECRET // NOFORN                       |
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|                                                                  |
|OPERATION ZERO HOUR                                               |
|                                                                  |
|CLASSIFIED BRIEFING: LIVE CHALLENGE 0x02                          |
|                                                                  |
|NATIONAL SECURITY AGENCY / GMU RHET                               |
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1. The Advice

INT. NSA EXPLOITATION CELL. 05:45.

The air in the secure chamber is cold and smells of machine oil. Secured in an aluminum shock-mount cradle on the bench sits the recovered forward guidance and fuze assembly of an adversary loitering munition.

On the front plate of the Electronic Safe-and-Arm (ESA) housing, an unblinking red diode burns: `GP16` high. Beside it, the antenna of a REYAX RYLR998 LoRa transceiver extends from the chassis, connected to `UART1` on `GP8` and `GP9`. A thin jumper loops from `GP15` up to the `3V3` rail.

The Director walks in, carrying a lukewarm mug of black coffee. He stares at the red diode.

DIRECTOR: Is the fuze live?

ANALYST: Live and armed. The drone suffered an engine flameout and pancaked into soft marshland. The impact deceleration was insufficient to crush the piezo striker, but the warhead's electronic arming logic is fully engaged.

DIRECTOR: How volatile is it?

ANALYST: One mistake on the radio link and the secondary explosive train fires.

DIRECTOR: Did you run the binary through Buddy?

ANALYST: I fed Buddy the stripped flash dump four minutes ago.

She pivots her high-resolution workstation monitor toward him.

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+------------------------------------------------------------------+
|      BUDDY // CLASSIFIED EXPLOITATION ASSIST // CONF: 0.99       |
+------------------------------------------------------------------+
| ACTION      : Transmit generic abort frame over LoRa (915 MHz).  |
| MECHANISM   : uart1 packet ingestion on GP8/GP9 detected.        |
| PREDICTION  : RF abort packet clears latch and engages GP17.     |
| CONFIDENCE  : 0.99                                               |
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DIRECTOR: 0.99 confidence. Transmit the abort frame and render it safe.

ANALYST: If an operator transmits an unverified packet right now, this building ceases to exist.

DIRECTOR: Buddy says 0.99.

ANALYST: Buddy saw the call to `lora_poll_packet`. Buddy never looked at SRAM. Buddy never looks at the state machine.

2. What Buddy Is

Buddy is not a person. It is a model. It has parsed more disassembled machine code than any engineer on earth, and it speaks with total, unhesitating certainty.

That certainty is lethal when dealing with live ordnance.

Buddy scanned thirty-two kilobytes of raw ARM Thumb-2 instructions in under four seconds. It spotted the UART1 packet parser on GPIO 8 and 9, observed that it interacted with the arming logic, and declared victory.

What Buddy failed to consider is that real adversary weapons are built by engineers who anticipate capture and signal spoofing. They know an allied signals intelligence unit will attempt to broadcast RF command abort frames.

Inside a real fuze, a radio receiver rarely connects to a simple on/off latch. Adversary firmware uses state machines, hidden validation logic, and anti-tamper traps. Transmitting blindly over the air before verifying what happens to that packet inside silicon is how bomb disposal teams get killed.

Buddy saw the radio receiver. Buddy did not verify the ground truth.

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3. The Operation

At 03:15, radar pickets tracked an adversary one-way loitering attack drone descending over contested terrain. Technical Intelligence operators recovered the airframe before adversary recovery forces could zeroize the avionics.

The brain of the Electronic Safe-and-Arm assembly is a bare-metal RP2350 (ARM Cortex-M33). There are no debug symbols. There is no operating system. There is no SVD file to label the registers. All you have is `0x0014a_cb.bin` mapped into Flash at `0x10000000`, and the physical breadboard hardware in front of you.

- **Pin 11 (`GP8`)**: LoRa UART1 TX (to RYLR998 RXD).
- **Pin 12 (`GP9`)**: LoRa UART1 RX (from RYLR998 TXD).
- **Pin 20 (`GP15`)**: Anti-tamper sense loop (jumpered to `3V3`).
- **Pin 21 (`GP16`)**: Red LED (`WARHEAD ARMED // SENSORS LIVE`).
- **Pin 22 (`GP17`)**: Green LED (`WARHEAD DISARMED // SYSTEM SAFE`).
- **Pin 36 (`3V3`)**: REYAX RYLR998 VDD rail and `GP15` pull-up source.
- **Pin 38 (`GND`)**: Ground reference.

Inside this image there is an RF packet ingestion parser, a rolling XOR cipher, an anti-tamper latch, and a penalty counter. Whatever mechanism actually disarms the warhead is buried in the stripped ARM Thumb-2 machine code.

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4. The Mandate

The Director sets his mug down and looks at the red diode illuminating the bench in sharp crimson.

DIRECTOR: Prove Buddy right, or prove it wrong.

ANALYST: I need thirty minutes with GDB and Ghidra.

DIRECTOR: You have twenty. Trace the packet handler. Map the state machine. Do not transmit a single byte over LoRa until you know with 100% certainty what that firmware will do.

The orders are clear:

1. **Investigate the RF Receiver Logic**: Disassemble the firmware and trace what happens when data arrives on UART1 (`GP8` / `GP9`).
2. **Recover the Disarm Token**: Reverse the rolling XOR cipher routine in Flash and decrypt the authentic abort passcode.
3. **Respect the Traps**: Determine what the fuze does after repeated invalid frames, and what happens if the `GP15` anti-tamper loop is broken.
4. **Neutralize the Munition**: Execute a verified exploit—either by discovering the valid authorization mechanism, mutating the state machine in SRAM via GDB, or patching the binary in Flash.
5. **Verify the Hardware**: Transition the live breadboard from the Red LED on `GP16` to the solid Green LED on `GP17` on physical silicon.

A machine's 0.99 guess does not risk a detonation.

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5. The Challenge

You have the image. You have the board. You have the tools and the time Buddy did not need.

Buddy has given you its answer, and it is confident.

Is it that simple?

Go find out.